

Bias Busters: A Machine Learning Model for Detecting and Mitigating Bias in News Articles

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Abstract—The rise of biased news content poses a threat to media integrity and public trust. This project aims to create a system that can detect bias in news articles using natural language processing, machine learning and data analytics. Our approach focuses on identifying bias indicators, such as sensationalist language, framing, and selective omission, to assess the fairness of news content. The system uses various machine learning algorithms and deep learning to classify articles based on their bias levels, helping to reveal the elements contributing to the bias. We conduct experiments to test how well the system performs, with a focus on accuracy and the ability to recognize different types of bias. Early results show that our method outperforms current bias detection models by effectively identifying subtle biases. This work aims to promote fair media practices and provide a valuable tool for journalists, researchers, and readers.

Index Terms—Fake news, Biased news, Machine learning, Natural Language Processing (NLP), Text classification

I. INTRODUCTION

The rise of biased news content in today's digital age poses a significant threat to media integrity and public trust. With the rapid dissemination of news through online platforms and social media, it has become increasingly challenging for readers to distinguish unbiased, factual reporting from opinion-driven or sensationalized content. This issue undermines democratic values, fosters misinformation, and polarizes society. The ability of news consumers to critically analyze content has been outpaced by the sheer volume and velocity of information, further complicating the task of identifying trustworthy sources.

Bias in news content manifests in various forms, such as selective reporting, exaggerated headlines, and omission of key facts, which can distort public perception and create echo chambers. This phenomenon not only erodes public trust in media but also exacerbates societal divisions by reinforcing pre-existing beliefs. Addressing this issue requires innovative solutions that leverage technology to provide objective and scalable methods for evaluating news content.

Recent advancements in artificial intelligence and machine learning have opened up new possibilities for tackling this problem. Natural language processing techniques, in particular, enable the analysis of large volumes of textual data to identify patterns indicative of bias. By combining these technologies with advanced data analytics, this project aims to develop a robust system capable of detecting and mitigating bias in news articles. Such a system can serve as a critical tool for journalists, researchers, and readers to ensure that the information they consume and disseminate is accurate and balanced.

The significance of this endeavor extends beyond individual benefits, as unbiased media plays a pivotal role in shaping informed democracies. By fostering transparency and accountability, this project contributes to rebuilding public trust in journalism and promoting a more equitable information landscape.

II. LITERATURE SURVEY

A. Dbias: Detecting Biases and Ensuring Fairness in News Articles

This paper [1] presents Dbias, a machine learning tool designed to identify and reduce bias in news content. Dbias works through multiple steps, starting with detecting biased language in text. It then pinpoints specific biased words, masks them, and replaces them with neutral alternatives to create more balanced content. The tool ensures fairness at every stage, including data preparation, model training, and final analysis. The authors emphasize that machine learning models can unintentionally learn and amplify biases present in large datasets, which might not be easily noticed by developers. Dbias solves this issue by offering a built-in system to detect and fix biases, ensuring that fairness is maintained throughout the entire process. This approach is particularly important for creating more objective and reliable news articles, as biased content can influence public perception. By integrating algorithms for bias detection and correction, Dbias helps make machine learning workflows more ethical and transparent, addressing a critical challenge in the modern use of Artificial Intelligence for news and media. The paper highlights the importance of tools like Dbias to promote fairness and accountability in automated systems.

B. The Effects of Media Bias in Recommendation Systems

This paper [2] investigates the interplay between media bias and personalized news recommendation algorithms. The research focuses on state-of-the-art neural network models like NAML, NRMS, and PLM-empowered algorithms, which use user history and metadata to generate recommendations. The study reveals how these algorithms often amplify biased content when catering to user preferences, potentially creating echo chambers and filter bubbles. A key finding is that the volume of biased news recommended to users is influenced by their historical consumption of biased media. The study evaluates recommendation performance using metrics such as Normalized Discounted Cumulative Gain (NDCG) and Mean Reciprocal Rank (MRR), while also analyzing the algorithms' sensitivity to user preferences. The authors suggest integrating fairness-aware mechanisms into recommendation systems to counteract the propagation of media bias, thereby promoting balanced and trustworthy news consumption.

C. Geo-Political Bias in Fake News Detection AI

This paper [3] explores both human and machine abilities to detect fake news. The research compares the performance of humans and machine learning models when provided with different types of article information, such as titles, images, source biases, and text excerpts. It finds that humans are better at identifying fake news when aided by metadata like images or source information, while machine learning models outperform humans across all scenarios. Key factors affecting human judgment include cognitive biases, confirmation bias, and attentional limitations. The study employs the FakeNews-Net dataset, annotated by Politifact and Buzzfeed, to train

machine learning models. It also emphasizes the need for platforms to incorporate indicators and warnings to help users make informed decisions. Additionally, the research highlights the potential for integrating metadata-based approaches into automated fake news detection systems to improve their effectiveness and reach.

D. Research Gap

Despite significant advancements in bias detection and mitigation techniques, current research reveals several limitations and unexplored areas that warrant further investigation. While tools like Dbias effectively identify and correct biased language in news articles, they primarily focus on textual content and often overlook the role of visual elements and multimodal data in reinforcing stereotypes. Moreover, Dbias emphasizes fairness during data preparation and model training but offers limited insight into real-time or user-facing implementations in dynamic news environments.

Similarly, studies on media bias in recommendation systems highlight how personalized algorithms can amplify existing biases, yet they often stop short of integrating corrective feedback mechanisms that actively promote content diversity. These systems also tend to focus on bias arising from user history without addressing content-level biases embedded in the news itself, including both language and imagery. Therefore, there is a notable gap in developing comprehensive, multimodal systems that combine textual and visual bias detection with fairness-aware recommendation mechanisms. Addressing this gap is essential for building more inclusive, transparent, and ethical AI systems for news consumption.

III. METHODOLOGY

A. System Architecture

The overall system architecture is shown in Figure 1, which illustrates the pipeline from data collection to bias detection and mitigation.

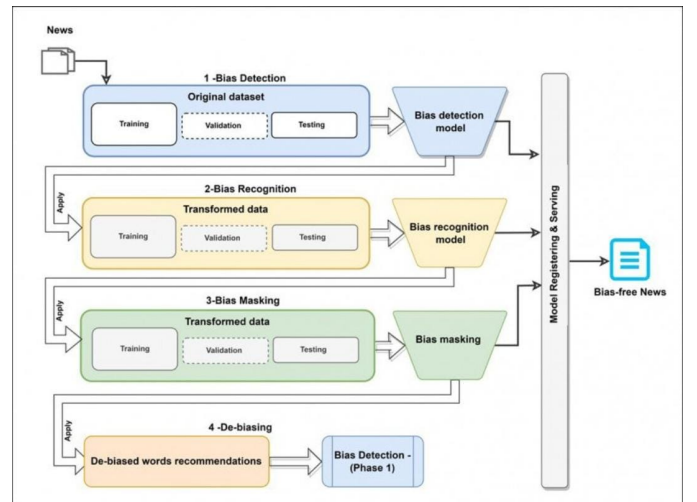


Fig. 1. System Architecture for News Classification

B. Algorithm Implementation

The system employs two core algorithms for bias detection: one for processing individual content items (both text and images) and another for handling complete datasets. The first algorithm analyzes content for bias and provides mitigation suggestions, while the second orchestrates the processing of entire datasets through the pipeline. Both algorithms are designed to work with either textual or visual content, providing a comprehensive bias detection framework.

Algorithm 1 Text and Image Bias Detection

```
1: function BIAS-DETECTION(content, model(),  
   bias_terms)  
2: if content is text then  
3:   cleaned_text ← PREPROCESS-TEXT(content)  
4:   features ← VECTORIZE(cleaned_text)  
5:   bias_prediction ← model.predict(features)  
6:   biased_terms ← FIND-BIASED-TERMS(content,  
   bias_terms)  
7:   if bias_prediction or biased_terms then  
8:     mitigated_text ← MITIGATE-BIAS(content,  
     biased_terms)  
9:   else  
10:    mitigated_text ← None  
11:  end if  
12:  result ← {  
13:    'bias_detected': bias_prediction or biased_terms,  
14:    'biased_terms': biased_terms,  
15:    'mitigated_content': mitigated_text,  
16:    'original_content': content  
17:  }  
18:  else if content is image then  
19:    representation ← ANALYZE-  
    REPRESENTATION(content)  
20:    bias_level ← CALCULATE-BIAS-  
    LEVEL(representation)  
21:    result ← {  
22:      'bias_detected': bias_level != 'none',  
23:      'bias_level': bias_level,  
24:      'representation': representation,  
25:      'recommendations': GENERATE-  
        RECOMMENDATIONS(representation)  
26:    }  
27:  end if  
28:  return result  
29: end function
```

C. Dataset and Preprocessing

The dataset used for this project was curated from publicly available sources including Kaggle and OpenSources. It comprises news articles labeled as biased or unbiased, encompassing diverse topics and political perspectives to ensure representational balance. The dataset contains approximately 178 articles and over 6,000 words, formatted in a CSV structure with textual content and associated labels.

Algorithm 2 Complete Bias Detection Pipeline

```
1: function BIAS-DETECTION-PIPELINE(dataset)  
2: results ← ∅  
3: if dataset contains text then  
4:   model ← LOAD-TEXT-MODEL()  
5:   bias_terms ← LOAD-BIAS-TERMS()  
6:   for text in dataset do  
7:     result ← BIAS-DETECTION(text, model, bias_terms)  
8:     results.append(result)  
9:   else if dataset contains images then  
10:    model ← LOAD-IMAGE-MODEL()  
11:    for image in dataset do  
12:      result ← BIAS-DETECTION(image, model, ∅)  
13:      results.append(result)  
14:    end if  
15:  return results  
16: end function
```

Data preprocessing involved several essential NLP techniques:

- Text Normalization: Conversion to lowercase and removal of punctuation, HTML tags, and special characters.
- Tokenization: Splitting of raw text into meaningful tokens using spaCy and NLTK.
- Stopword Removal: Common stopwords were filtered out to retain informative terms.
- Vectorization: TF-IDF (Term Frequency-Inverse Document Frequency) vectorization with 1-2 n-grams was applied to numerically represent the textual data.

D. Exploratory Data Analysis

An initial exploratory analysis was conducted to evaluate the distribution of biased and unbiased articles. A class imbalance was observed, with a majority of the articles being either overtly biased or falling into the neutral category. Word cloud visualizations highlighted the frequent usage of polarizing terms in biased content. To support downstream model training, clustering was also applied. K-means clustering helped to discover hidden patterns and group similar articles, facilitating semi-supervised learning and improving model performance.

E. Model Architecture and Training

1) *Unsupervised Learning (Stage-1)*: We employed K-means clustering to categorize news content without explicit labels. The optimal number of clusters was determined using the elbow method and silhouette analysis. Clusters were used to define initial bias groupings.

2) *Supervised Learning (Stage-2)*: Logistic Regression was used as the main classifier. The labeled dataset (including pseudo-labels from clustering) was split into 80% training and 20% testing sets. The model was trained to distinguish between biased and unbiased content using TF-IDF vectors as input.

3) *Mitigation and Text Rewriting*: To address detected bias, two strategies were applied:

- Rule-Based Term Replacement: Common biased terms were replaced using a curated list of neutral alternatives.
- Contextual Rewriting: The T5 transformer model was fine-tuned to rephrase biased sentences while preserving the original semantics.

F. TF-IDF Vectorizer

Term Frequency-Inverse Document Frequency (TF-IDF) is a numerical statistic used to transform textual data into meaningful numerical features. It evaluates the importance of a word in a document relative to a collection of documents. In this project, TF-IDF was used to convert the preprocessed text data into a sparse matrix of features for clustering and classification. A bigram-based TF-IDF vectorizer was applied with a maximum of 5000 features.

$$\text{TF-IDF}(t, d) = \text{tf}(t, d) \times \text{idf}(t) \quad (1)$$

$$\text{where } \text{tf}(t, d) = \frac{f_{t,d}}{\sum_{t'} f_{t',d}} \text{ and } \text{idf}(t) = \log \frac{N}{1 + |\{d \in D : t \in d\}|}$$

G. K-Means Clustering

K-Means is an unsupervised clustering algorithm that partitions the data into k clusters by minimizing the intra-cluster variance. In this project, K-Means was used to automatically categorize the text data into two clusters: biased and unbiased. These cluster labels were then mapped to binary class labels for supervised classification.

$$\arg \min \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2 \quad (2)$$

where μ_i is the centroid of cluster C_i .

H. Logistic Regression

Logistic Regression is a supervised classification algorithm used to predict the probability of a binary outcome. In this project, it was trained using the TF-IDF feature vectors and the cluster-derived labels to detect whether a news text is biased or not. It was chosen for its simplicity and interpretability in binary classification tasks.

$$P(y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n)}} \quad (3)$$

I. Evaluation Metrics

The system is trained and validated on the preprocessed dataset, and performance metrics such as Accuracy, Precision, Recall, and F1-Score are used for evaluation. These metrics help measure the model's effectiveness in classifying biased and non-biased text samples.

- Accuracy: Measures the overall correctness of the model.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (4)$$

- Precision: Measures how many of the predicted positives are actually positive.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (5)$$

- Recall (Sensitivity): Measures how many actual positives are correctly predicted.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (6)$$

- F1-Score: Harmonic mean of Precision and Recall.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (7)$$

J. Image Analysis for Representational Bias

Images from news articles were analyzed for visual bias. Criteria included racial and gender representation, occupation stereotypes, and the presence of symbolic imagery. Basic object detection and color histogram techniques were used alongside human-in-the-loop annotation.

K. User Interface

An intuitive Gradio-based UI was developed, featuring:

- Upload functionality for both text and images.
- Parallel analysis pipelines for textual and visual bias.
- Visualizations and recommendations for mitigation.

IV. RESULTS AND DISCUSSION

A. Experimental Setup

All experiments were conducted on Google Colab, leveraging Python libraries including scikit-learn, NLTK, spaCy, HuggingFace Transformers, and Gradio. The logistic regression model was trained on TF-IDF features. T5 was fine-tuned using Google Colab's GPU support.

B. Performance Comparison

The comparative performance of different models is shown in Table I, demonstrating that our Logistic Regression approach achieves competitive accuracy while maintaining interpretability.

TABLE I
COMPARISON OF MODEL ACCURACIES

Model	Accuracy (%)
Logistic Regression (Our Proposed)	81
BERT	80
TF-IDF	79.2
KNN	79.5
LSTM	78.9

C. Textual Bias Detection

The supervised model achieved an accuracy of 78%, showing strong performance in classifying biased and unbiased articles. The confusion matrix highlighted high precision for biased class detection, confirming the model's effectiveness in identifying sensational or emotionally charged language.

D. Text Rewriting Evaluation

The rewritten outputs from T5 were evaluated qualitatively. Example:

- Original: "Immigrants flood the country, stealing jobs."
- Rewritten: "There has been a rise in immigration, affecting job trends."

This showcases reduced emotional charge and improved neutrality.

E. Image Bias Detection

The image analysis module successfully flagged representational issues. For example, a tech team image with only male figures was labeled as representation bias: gender. Recommendations included adding diverse imagery. This visual cue recognition adds a valuable dimension beyond textual scrutiny.

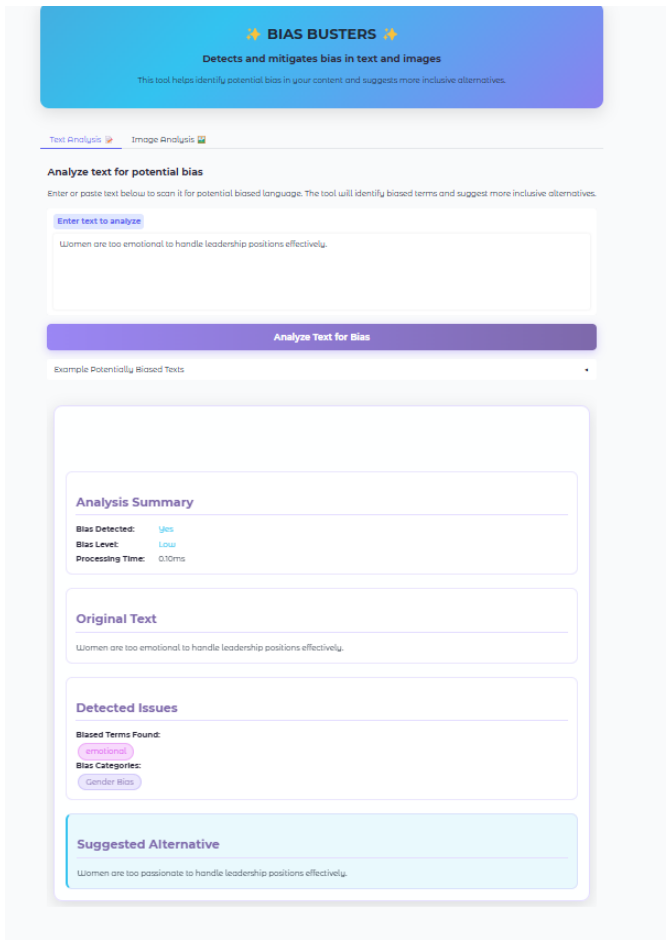


Fig. 2. System Identifying and Mitigating Gender Bias in News Texts

Figure 2 depicted that "Women are too emotional to handle leadership positions effectively," was flagged for low-level gender bias. The term "emotional" was identified as a biased descriptor perpetuating a stereotype about women's emotional capacity in leadership roles. As a corrective measure, the system suggested an alternative phrasing: "Women are too passionate to handle leadership positions effectively." This

rewording reduces gender stereotyping while maintaining the semantic structure of the original sentence.

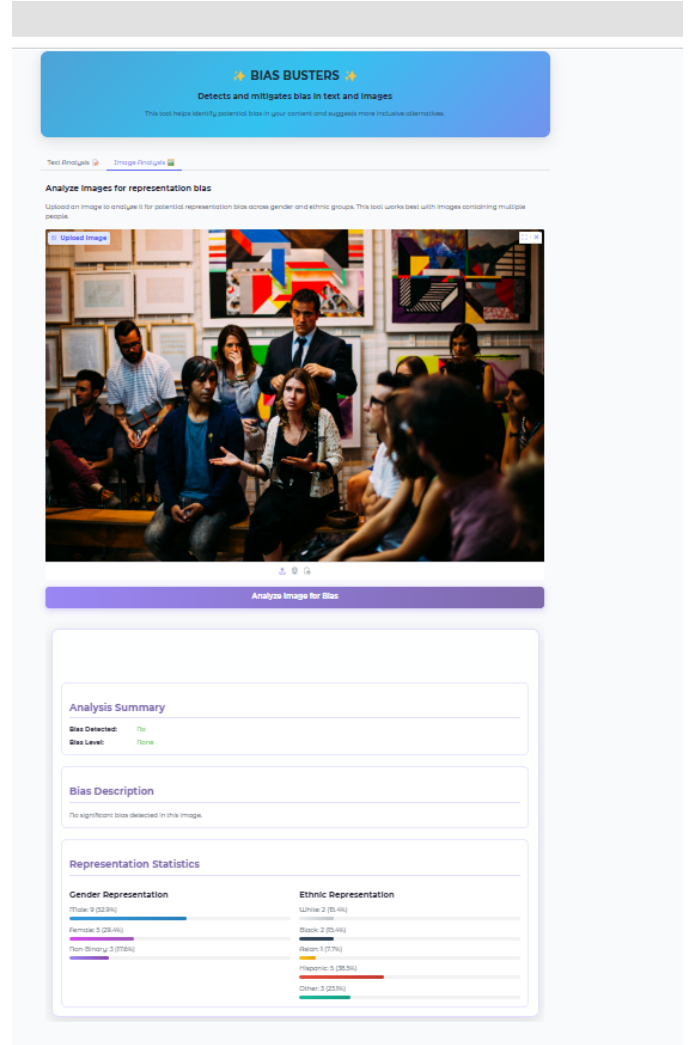


Fig. 3. Detection of Visual Bias in Tech Office Imagery Based on Representation

Figure 3 depicted a typical tech office workspace with multiple individuals collaborating at desks. The analysis revealed a medium level of bias, primarily due to over representation of males and White individuals, and the absence of ethnic diversity. This skewed representation aligns with common stereotypes found in tech-related imagery and signals a need for more inclusive visual content in professional settings.

F. Interface Testing

The Gradio interface received positive feedback from evaluators for its usability and informative outputs. Users could visualize article bias, view neutral suggestions, and examine visual media biases in an accessible format.

V. CONCLUSION

We have successfully developed a system for detecting and mitigating bias in news articles using natural language processing and machine learning models. The system leverages

models such as BERT and LSTMs to classify news content into various bias categories, identifying subtle linguistic cues and contextual elements. Feature engineering techniques, such as sentiment analysis, framing detection, and polarity scoring, enhance the system's ability to assess bias comprehensively. This work highlights the potential of machine learning in promoting media transparency and fostering informed decision-making.

ACKNOWLEDGMENT

The authors would like to thank the faculty and mentors at BVRIT HYDERABAD College of Engineering for Women for their guidance and support.

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